

Enhanced Unified Implementation Plan for Autonomous CNC Toolpath System

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June 8, 2026

1 Source Prompts

Toolpath Prompt

Generate toolpaths for fused cuboids avoiding vertical walls and entering from outside.

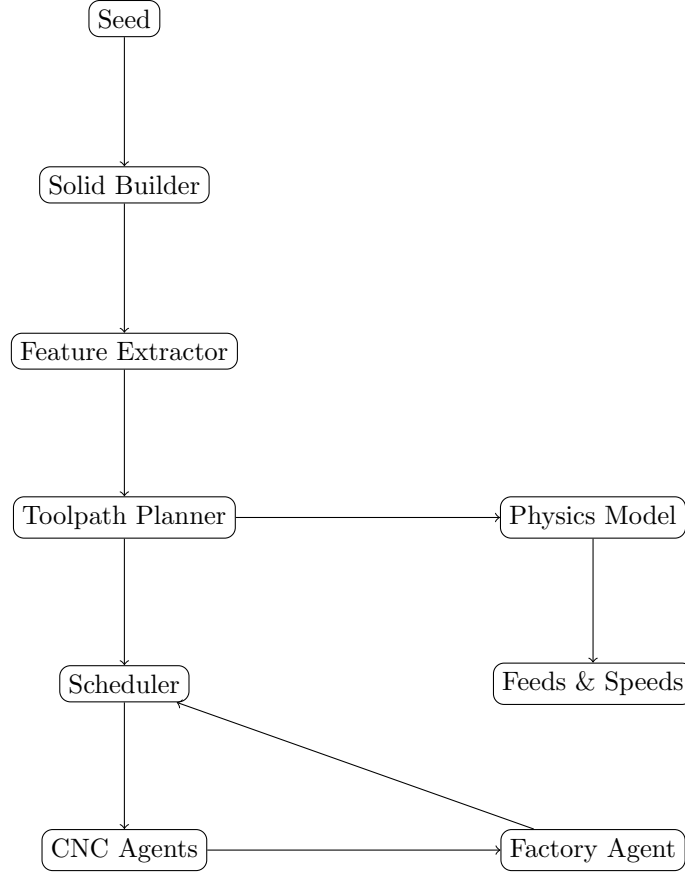
Feeds and Speeds Prompt

Select feeds and speeds using practical machining heuristics.

Factory Simulation Prompt

Develop an autonomous multi-machine CNC factory with scheduling, tool tracking, and optimization.

2 System Architecture



3 Physics Model Integration

Cutting force model:

$$F_c = K_c \cdot A_c$$

SI: F_c in N, K_c (specific cutting force) in MPa, A_c chip area (mm²)

Power model:

$$P = \frac{F_c \cdot V}{60}$$

where P in kW, V in m/min

Torque model:

$$T = \frac{9550P}{N}$$

where T in Nm

Constraint: ensure P and T are below machine limits.

4 Mathematical Optimization

Multi-objective cost function:

$$J = w_1 T_m + w_2 C + w_3 \frac{1}{L} + w_4 S$$

Where:

- T_m : machining time
- C : cost
- L : tool life
- S : surface finish penalty

5 Tool Crib Management

- Tool crib resides at factory agent - Every $\Delta_t = 500$ steps:

- Query all machines
- Identify tools with $< 20\%$ life
- Schedule replacement at part change

Critical rule:

- If tool life $< 10\% \rightarrow$ immediate machine stop

6 Global Timer System

Define synchronized clock:

$$t_{real} = n \cdot t_{unit}$$

Default: $t_{unit} = 60s$

All agents operate on discrete time steps ensuring concurrency.

7 Error Handling Architecture

7.1 CNC Agent

- Dashboard with live toolpath - User prompt box triggers error - Convert to structured context - Send to local LLM

Catastrophic condition:

- Stop machine
- Send state to factory agent

7.2 Factory Agent

- Receives error + state - Converts to LLM prompt - Evaluates responses - Selects mitigation strategy - Sends commands to CNC agent

8 Improved Modules

8.1 Toolpath Planner

- Safe region computation - Entry strategies (ramp, helix)

8.2 Physics-Constrained Feeds

- Limit feed by power/force

8.3 Scheduler

- Uses timer - Coordinates machine states

9 Remaining Weaknesses

- No full chatter stability model
- Limited geometry complexity
- LLM reliability for error recovery unknown
- No real-time sensor feedback

10 Conclusion

This improved plan integrates physics, optimization, concurrency, and robust error handling to create a realistic autonomous CNC system.